

Lead Concentration Analysis Report

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1 Descriptive Statistics

The analysis was performed on a dataset containing lead concentration measurements and related variables.

1.1 Key Variables

- **MAXFWT**: Maximum finger-wrist tapping score
- **Ld72**: Lead concentration measurement 1
- **Ld73**: Lead concentration measurement 2
- **Sex**: Gender (male/female)
- **Age**: Age of participants
- **Iqf**: Intelligence quotient
- **Totyrs**: Total years

1.2 Summary Statistics

1.2.1 Numeric Variables Summary

- **MAXFWT**:
 - Mean: 52.05
 - Median: 52.00
 - Min: 13.00
 - Max: 84.00
 - Q1: 47.00
 - Q3: 59.00
- **Ld72**:
 - Mean: 34.17
 - Median: 31.00
 - Min: 1.00

- Max: 99.00

- Q1: 24.00

- Q3: 38.00

- **Ld73:**

- Mean: 31.58

- Median: 28.00

- Min: 15.00

- Max: 58.00

- Q1: 24.00

- Q3: 38.00

- **Iqf (Intelligence Quotient):**

- Mean: 91.91

- Median: 91.50

- Min: 50.00

- Max: 141.00

- Q1: 82.25

- Q3: 100.75

- **Age:**

- Mean: 9.05

- Median: 8.67

- Min: 3.75

- Max: 15.92

- Q1: 6.38

- Q3: 12.06

- **Total Years (Totyrs):**

- Mean: 6.73
- Median: 6.00
- Min: 1.00
- Max: 15.00
- Q1: 4.00
- Q3: 9.00

1.2.2 Gender Distribution

- Male: 63 (61.8%)
- Female: 39 (38.2%)
- Total Sample Size: 102

2 Outlier Analysis

The following outliers were detected using the $1.5 \times \text{IQR}$ rule:

- **MAXFWT (Maximum finger-wrist tapping score):**

- Outliers found at observations: 17, 30, 34, 49, 50, 73, 75
- These represent 7 out of 102 observations (6.9% of the data)

- **Ld72 (Lead concentration measurement 1):**

- Outliers found at observations: 5, 26, 33, 64, 69, 72, 78, 81, 82, 87
- These represent 10 out of 102 observations (9.8% of the data)

- **Ld73 (Lead concentration measurement 2):**

- No outliers detected

- **Age:**

- No outliers detected

Note: The presence of outliers in MAXFWT and Ld72 may indicate:

1. Extreme lead exposure cases
2. Potential measurement errors
3. Natural variation in the population

These outliers should be considered in the context of their biological plausibility before making decisions about their treatment in the analysis.

3 Normality and Homoscedasticity Tests

3.1 Normality Test Results (p-values)

- **MAXFWT:**
 - Shapiro-Wilk test: $p < 0.001$
 - Anderson-Darling test: $p < 0.001$
 - Conclusion: Non-normal distribution
- **Ld72:**
 - Shapiro-Wilk test: $p < 0.0001$
 - Anderson-Darling test: $p < 0.0001$
 - Conclusion: Non-normal distribution
- **Ld73:**
 - Shapiro-Wilk test: $p < 0.0001$
 - Anderson-Darling test: $p < 0.0001$
 - Conclusion: Non-normal distribution
- **Age:**
 - Shapiro-Wilk test: $p < 0.001$
 - Anderson-Darling test: $p < 0.001$
 - Conclusion: Non-normal distribution

3.2 Homoscedasticity Tests for MAXFWT by Sex

- Levene's Test: $p = 0.178$ (homoscedasticity cannot be rejected)
- Bartlett's Test: $p = 0.008$ (heteroscedasticity detected)

Note: All variables show significant deviation from normality ($p < 0.05$), suggesting that:

1. Non-parametric tests should be considered for analyses
2. Data transformations might be appropriate
3. The Bartlett's test indicates potential heteroscedasticity in MAXFWT between sexes, while Levene's test does not

4 Confidence Intervals for MAXFWT by Sex

- **90% Confidence Intervals:**
 - Male: 47.2 to 54.2 (Mean: 50.7, SE: 2.08)
 - Female: 51.4 to 56.9 (Mean: 54.1, SE: 1.63)
- **95% Confidence Intervals:**
 - Male: 46.5 to 54.9 (Mean: 50.7, SE: 2.08)
 - Female: 50.8 to 57.4 (Mean: 54.1, SE: 1.63)
- **99% Confidence Intervals:**
 - Male: 45.1 to 56.3 (Mean: 50.7, SE: 2.08)
 - Female: 49.7 to 58.6 (Mean: 54.1, SE: 1.63)

Note: The confidence intervals show that:

1. Female participants have higher MAXFWT scores on average
2. The intervals are narrower for females, indicating more precise estimates
3. As confidence level increases (90% to 99%), the intervals become wider to provide greater certainty
4. There is some overlap in the intervals, particularly at higher confidence

levels

4.1 Correlation Analysis

- Correlation between MAXFWT and Ld72: -0.266 (weak negative correlation)
- Correlation between MAXFWT and Ld73: -0.341 (moderate negative correlation)

Note: Both correlations are negative, indicating that as lead concentration increases, the maximum finger-wrist tapping score tends to decrease.

5 Hypothesis Testing

Three main hypothesis tests were performed:

5.1 Gender Difference in MAXFWT

Two Sample t-test Results:

- t-statistic: -1.1929
- p-value: 0.2364
- Degrees of freedom: 81
- 95% Confidence Interval: $[-9.181, 2.299]$
- Mean MAXFWT (Male): 50.68
- Mean MAXFWT (Female): 54.12
- Conclusion: No significant difference in MAXFWT between males and females ($p > 0.05$)

5.2 Lead Level Impact (Ld72)

Welch's T-test Results (High vs Low Ld72):

- t-statistic: -3.0115
- p-value: 0.9971
- Degrees of freedom: 25.61

- 95% Confidence Interval: $[-16.101, \infty]$
- Mean MAXFWT (High Ld72): 44.00
- Mean MAXFWT (Low Ld72): 54.28
- Conclusion: No significant difference in MAXFWT between high and low Ld72 groups

5.3 Interaction Analysis (Gender and Lead Type)

ANOVA Results:

- F-statistic: 2.72
- p-value: 0.0145
- Degrees of freedom: 7 (Group), 74 (Residuals)
- Significant at $\alpha = 0.05$ level

Post-hoc Tukey HSD Results (Significant Differences):

- Male Type4 vs Male Type1: $p = 0.005$ (significant)
- Other pairwise comparisons were not statistically significant

Note:

1. The gender difference test showed no significant difference in MAXFWT scores
2. The lead level impact test did not show significant differences between high and low Ld72 groups
3. The ANOVA revealed significant differences across gender and lead type combinations
4. Post-hoc analysis identified specific significant differences between certain groups

6 Linear Regression Analysis

Linear Regression Results (MAXFWT $\sim$ Ld73):

6.1 Model Summary

- R-squared: 0.1164
- Adjusted R-squared: 0.1055
- F-statistic: 10.67
- p-value: 0.0016
- Residual standard error: 12.2 on 81 degrees of freedom
- 19 observations deleted due to missingness

6.2 Coefficients

- Intercept: 64.3660 (SE: 4.0020, $p < 0.001$)
- Ld73 slope: -0.3938 (SE: 0.1206, $p = 0.0016$)

6.3 95% Confidence Intervals

- Ld73 slope: $[-0.634, -0.154]$

6.4 Residuals Summary

- Minimum: -42.308
- 1st Quartile: -5.808
- Median: 0.993
- 3rd Quartile: 7.584
- Maximum: 30.661

6.5 Predicted MAXFWT Reduction

- At Ld73 = 100 $\mu\text{g}/100\text{ ml}$: 39.38 units
- This represents the expected decrease in MAXFWT compared to Ld73 = 0

6.6 Interpretation

1. The model is statistically significant ($p = 0.0016$)
2. Ld73 has a significant negative effect on MAXFWT ($p = 0.0016$)

3. Each unit increase in Ld73 is associated with a 0.394 unit decrease in MAXFWT
4. The model explains approximately 11.6% of the variance in MAXFWT
5. The 95% confidence interval for the slope does not include zero, confirming the significance of the relationship

7 Key Findings

1. Gender differences in MAXFWT:

- No significant difference between males and females ($t = -1.1929$, $p = 0.2364$)
- Mean MAXFWT for males: 50.68, females: 54.12
- 95% CI for difference: $[-9.181, 2.299]$

2. Lead concentration impact:

- Significant negative correlation between MAXFWT and Ld73 ($r = -0.341$)
- Significant negative correlation between MAXFWT and Ld72 ($r = -0.266$)
- Higher lead levels associated with lower MAXFWT scores
- Linear regression shows significant negative effect ($\beta = -0.394$, $p = 0.0016$)

3. Gender-Lead Type interaction:

- Significant overall effect ($F = 2.72$, $p = 0.0145$)
- Significant difference between Male Type4 and Male Type1 ($p = 0.005$)
- Other group comparisons showed no significant differences

4. Linear relationship:

- R-squared: 0.1164 (11.64% of variance explained)
- Significant negative slope ($\beta = -0.394$, $p = 0.0016$)
- 95% CI for slope: $[-0.634, -0.154]$
- Model is statistically significant ($F = 10.67$, $p = 0.0016$)

5. MAXFWT reduction:

- Estimated reduction of 39.38 units at $Ld73 = 100 \mu g/100 \text{ ml}$
- This represents a substantial decrease from baseline MAXFWT

8 Recommendations

1. Focus on lead exposure reduction:

- The strong negative correlation between lead levels and MAXFWT suggests that reducing lead exposure could improve motor function
- Target interventions for individuals with lead levels above 40 units

2. Monitoring and assessment:

- Implement regular MAXFWT assessments for individuals with elevated lead levels
- Pay particular attention to individuals in the highest lead exposure group (Type4)

3. Gender-specific considerations:

- While no significant gender differences were found in MAXFWT, maintain gender-balanced monitoring
- Consider gender-specific factors in lead exposure reduction strategies

4. Data collection improvements:

- Address the 19 missing observations in the dataset
- Consider collecting additional variables that might explain the remaining 88.36% of variance in MAXFWT

5. Further research:

- Investigate the mechanisms behind the significant interaction between gender and lead type
- Explore potential non-linear relationships between lead levels and MAXFWT
- Consider longitudinal studies to track changes in MAXFWT with lead

exposure over time